

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: Neural Network Representation, Perceptron Model, Multilayer Perceptron's, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks:25

Annexure A

Analytical Problem Solving

Code of Conduct:

I _M. Vidhya(192412151)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1. A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is -0.5 . For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails (5 Marks)

Sol)

Given

- Study Hours, $x_1 = 2$
- Attendance, $x_2 = 1$
- Weight for Study Hours, $w_1 = 0.6$
- Weight for Attendance, $w_2 = 0.4$
- Bias, $b = -0.5$

Step 1: Formula for Net Input

The net input of a perceptron is calculated as:

$$\boxed{\text{Net Input} = w_1x_1 + w_2x_2 + b}$$

Step 2: Substitute the Given Values

$$\text{Net Input} = (0.6 \times 2) + (0.4 \times 1) + (-0.5)$$

Step 3: Calculate Each Term

Calculate the first term:

$$0.6 \times 2 = 1.2$$

Calculate the second term:

$$0.4 \times 1 = 0.4$$

Add the bias:

$$\begin{aligned}\text{Net Input} &= 1.2 + 0.4 - 0.5 \\ &= 1.6 - 0.5 \\ &= 1.1\end{aligned}$$

Therefore,

$$\boxed{\text{Net Input} = 1.1}$$

Step 4: Apply the Step Activation Function

The step activation function is:

$$Y = \begin{cases} 1, & \text{if Net Input} \geq 0 \\ 0, & \text{if Net Input} < 0 \end{cases}$$

Since,

$$1.1 \geq 0$$

the output is:

$$\boxed{Y = 1}$$

Step 5: Conclusion

The output of the perceptron is 1, which represents Pass.

Final Answer

- Net Input = 1.1
- Output (Step Activation Function) = 1
- Conclusion: The student passes.

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)

Sol)

Given

- Initial Weight, $w = 0.5$
- Learning Rate, $\eta = 0.1$
- Input, $x = 2$
- Actual Output, $O = 0.6$
- Target Output, $T = 1$

Step 1: Calculate the Error

Formula

$$\text{Error} = \text{Target Output} - \text{Actual Output}$$

Substitute the Given Values

$$\text{Error} = 1 - 0.6$$

Calculation

$$\text{Error} = 0.4$$

Therefore,

$$\text{Error} = 0.4$$

Step 2: Calculate the Weight Update (Delta Rule)

Formula

$$\Delta w = \eta \times \text{Error} \times \text{Input}$$

where:

- η = Learning Rate
- Error = Target – Actual Output
- Input = Input value

Substitute the Given Values

$$\Delta w = 0.1 \times 0.4 \times 2$$

Calculation

First,

$$0.4 \times 2 = 0.8$$

Then,

$$0.1 \times 0.8 = 0.08$$

So,

$$\Delta w = 0.08$$

Step 3: Calculate the New Updated Weight

Formula

$$\text{New Weight} = \text{Old Weight} + \Delta w$$

Substitute the Given Values

$$\text{New Weight} = 0.5 + 0.08$$

Calculation

$$\text{New Weight} = 0.58$$

Therefore,

$$\text{New Weight} = 0.58$$

Final Answer

- Error = 0.4
- Weight Update (Δw) = 0.08
- New Updated Weight = 0.58

Conclusion

After one iteration of backpropagation using the delta learning rule, the neuron's weight is updated from 0.5 to 0.58, reducing the error and improving the learning process.

3.A perceptron model is used to classify whether a loan is approved based on two features: income and credit score. The weights are 0.7 and 0.5, and the bias is -0.6 . For an applicant with income = 1.5 and credit score = 2, compute the net input and apply the step activation function. Determine whether the loan is approved or rejected. (5 Marks)

Sol)

Given

- Income, $x_1 = 1.5$
- Credit Score, $x_2 = 2$
- Weight for Income, $w_1 = 0.7$
- Weight for Credit Score, $w_2 = 0.5$
- Bias, $b = -0.6$

Step 1: Formula for Net Input

The net input of the perceptron is calculated as:

$$\boxed{\text{Net Input} = w_1x_1 + w_2x_2 + b}$$

Step 2: Substitute the Given Values

$$\text{Net Input} = (0.7 \times 1.5) + (0.5 \times 2) + (-0.6)$$

Step 3: Calculate Each Term

First,

$$0.7 \times 1.5 = 1.05$$

Next,

$$0.5 \times 2 = 1.0$$

Now add the bias:

$$\begin{aligned}\text{Net Input} &= 1.05 + 1.0 - 0.6 \\ &= 2.05 - 0.6 \\ &= 1.45\end{aligned}$$

Therefore,

$$\boxed{\text{Net Input} = 1.45}$$

Step 4: Apply the Step Activation Function

The step activation function is:

$$Y = \begin{cases} 1, & \text{if Net Input} \geq 0 \\ 0, & \text{if Net Input} < 0 \end{cases}$$

Since,

$$1.45 \geq 0$$

the output is:

$$\boxed{Y = 1}$$

Step 5: Conclusion

The output of the perceptron is 1, which represents Loan Approved.

Final Answer

- Net Input = 1.45
- Output (Step Activation Function) = 1
- Conclusion: The loan is approved.

4. A classification model produces the following confusion matrix values: True Positives = 120, True Negatives = 200, False Positives = 30, and False Negatives = 50. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on these results, evaluate whether the model is suitable for applications where false negatives are critical.

Sol)

Given

- True Positives (TP) = **120**
- True Negatives (TN) = **200**

- False Positives (FP) = **30**
- False Negatives (FN) = **50**

Total number of samples:

$$\begin{aligned}
 & TP + TN + FP + FN \\
 &= 120 + 200 + 30 + 50 \\
 &= 400
 \end{aligned}$$

Step 1: Calculate Accuracy

Formula

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

Substitute the Given Values

$$= \frac{120 + 200}{120 + 200 + 30 + 50} \times 100$$

Calculation

$$\begin{aligned}
 &= \frac{320}{400} \times 100 \\
 &= 0.8 \times 100 \\
 &= \boxed{80\%}
 \end{aligned}$$

Step 2: Calculate Precision

Formula

$$\text{Precision} = \frac{TP}{TP + FP} \times 100$$

Substitute the Given Values

$$= \frac{120}{120 + 30} \times 100$$

Calculation

$$\begin{aligned}
 &= \frac{120}{150} \times 100 \\
 &= 0.8 \times 100
 \end{aligned}$$

$$= \boxed{80\%}$$

Step 3: Calculate Recall (Sensitivity)

Formula

$$\text{Recall} = \frac{TP}{TP + FN} \times 100$$

Substitute the Given Values

$$= \frac{120}{120 + 50} \times 100$$

Calculation

$$\begin{aligned} &= \frac{120}{170} \times 100 \\ &= 70.59\% \end{aligned}$$

$$\boxed{\text{Recall} = 70.59\%}$$

Step 4: Calculate Specificity

Formula

$$\text{Specificity} = \frac{TN}{TN + FP} \times 100$$

Substitute the Given Values

$$= \frac{200}{200 + 30} \times 100$$

Calculation

$$\begin{aligned} &= \frac{200}{230} \times 100 \\ &= 86.96\% \end{aligned}$$

$$\boxed{\text{Specificity} = 86.96\%}$$

Step 5: Calculate F1-Score

Formula

$$\text{F1-Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

Convert Precision and Recall into decimal values:

- Precision = **0.80**
- Recall = **0.7059**

Substitute the Values

$$F1 = \frac{2 \times (0.80 \times 0.7059)}{0.80 + 0.7059}$$

Calculation

$$\begin{aligned} &= \frac{2 \times 0.5647}{1.5059} \\ &= \frac{1.1294}{1.5059} \\ &= 0.75 \end{aligned}$$

Convert into percentage:

$$0.75 \times 100 = 75\%$$

$$\boxed{\text{F1-Score} = 75\%}$$

Step 6: Evaluation

The model has:

- Accuracy = 80%
- Precision = 80%
- Recall (Sensitivity) = 70.59%
- Specificity = 86.96%
- F1-Score = 75%

The **recall is only 70.59%**, meaning the model misses some positive cases (**50 false negatives**).

For applications where **false negatives are critical** (such as disease diagnosis or fraud detection), this model is **not fully suitable** because missing positive cases can lead to serious consequences. Improving **recall (sensitivity)** would make the model more reliable.

Final Answer

- Accuracy = 80%
- Precision = 80%
- Recall (Sensitivity) = 70.59%
- Specificity = 86.96%
- F1-Score = 75%

Conclusion: The model has good overall performance, but due to its relatively lower recall and the presence of false negatives, it is **not ideal for applications where false negatives are critical**. Improving recall would improve its suitability.

5. A Genetic Algorithm is applied to maximize the function $f(x) = 3x + 2$ for an initial population {2, 5, 7, 9}. Compute the fitness value for each individual and select the best candidates. Perform a crossover operation to generate new offspring and explain how mutation helps in maintaining diversity and avoiding local optima.
(5 Marks)

Sol)

Given

- **Fitness Function:**

$$f(x) = 3x + 2$$

- **Initial Population:**

$$\{2, 5, 7, 9\}$$

Step 1: Calculate the Fitness Value

Formula

$$f(x) = 3x + 2$$

For $x = 2$

$$\begin{aligned} f(2) &= 3(2) + 2 \\ &= 6 + 2 \\ &= 8 \end{aligned}$$

For $x = 5$

$$\begin{aligned} f(5) &= 3(5) + 2 \\ &= 15 + 2 \\ &= 17 \end{aligned}$$

For $x = 7$

$$\begin{aligned} f(7) &= 3(7) + 2 \\ &= 21 + 2 \\ &= 23 \end{aligned}$$

For $x = 9$

$$\begin{aligned} f(9) &= 3(9) + 2 \\ &= 27 + 2 \\ &= 29 \end{aligned}$$

Step 2: Fitness Table

Individual (x) Fitness $f(x) = 3x + 2$

2	8
5	17
7	23
9	29

Step 3: Select the Best Candidates

The individuals with the **highest fitness values** are:

- **$9 \rightarrow \text{Fitness} = 29$**
- **$7 \rightarrow \text{Fitness} = 23$**

Selected Parents: 9 and 7

Step 4: Perform Crossover

Binary Representation

$$9 = 1001$$

7 = 0111

Perform **single-point crossover** after the **second bit**.

Parents

Parent 1 : 10 | 01

Parent 2 : 01 | 11

Exchange the bits after the crossover point.

Offspring 1

10 | 11 = 1011 = 11

Offspring 2

01 | 01 = 0101 = 5

Step 5: Explain Mutation

Mutation is the process of randomly changing one or more bits in a chromosome.

Example

Chromosome:

1011 (11)

After mutation:

1010 (10)

Importance of Mutation

- Maintains diversity in the population.
- Prevents premature convergence.
- Helps escape local optima.
- Increases the chance of finding the global optimum.

Final Answer

Fitness Values

- $f(2) = 8$
- $f(5) = 17$
- $f(7) = 23$
- $f(9) = 29$

Best Candidates

- 9 (Fitness = 29)
- 7 (Fitness = 23)

Offspring after Crossover

- Offspring 1 = 11
- Offspring 2 = 5

Conclusion

Mutation introduces random changes in chromosomes, maintains genetic diversity, prevents the algorithm from getting stuck in local optima, and improves the search for the optimal solution.